

Continuous Convergence and Curried Functional Spaces

1. Preliminary definitions

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ generated by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.

If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x .

If $f : X \rightarrow Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X , then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \rightarrow x$, if it contains all open neighborhoods of x .

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter

$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$

converges to x . This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.

- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \emptyset$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X .
- A function $f : X \rightarrow Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \rightarrow x \in X$ implies $f(F) \rightarrow f(x) \in Y$.

Definition 1.4. One may try to decouple the notion of convergent filters from topology. Let X be a set with a relation $\rightarrow \subset \mathcal{F}(X) \times X$. Then $(X, \rightarrow)$ is called a *convergence space* if

- For all $x \in X$, we have $[x] \rightarrow x$;
- If $F \rightarrow x$ and $G \rightarrow x$, then $F \cap G \rightarrow x$;
- If $F \rightarrow x$ and $F \subset G$, then $G \rightarrow x$.

Definition 1.5. A map $f : X \rightarrow Y$ between convergence spaces is said to be *continuous* if $f(F) \rightarrow f(x) \in Y$ whenever $F \rightarrow x \in X$.

Definition 1.6. Let X, Y be two convergence spaces. Then the product $X \times Y$ can be endowed with the following convergence structure: a filter $F \in \mathcal{F}(X \times Y)$ converges to $(x, y) \in X \times Y$ if and only if $p_X(F) \rightarrow x \in X$ and $p_Y(F) \rightarrow y \in Y$.

Definition 1.7. Let X, Y be convergence spaces. By $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y . Then $\mathcal{C}(X, Y)$ can be endowed with a convergence structure as follows. For a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$, we say that $F \rightarrow f \in \mathcal{C}(X, Y)$ if and only if $F(P) \rightarrow f(p)$ whenever $P \rightarrow p \in X$. Here, the filter $F(P)$ is based on

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

The resulting convergence structure on $\mathcal{C}(X, Y)$ is called the structure of *continuous convergence*.

Remark 1.2. Continuous convergence on $\mathcal{C}(X, Y)$ is the coarsest convergence structure that makes the *evaluation mapping*

$$\omega : \mathcal{C}(X, Y) \times X \rightarrow Y$$

continuous. Here, the product $\mathcal{C}(X, Y) \times X$ is endowed with the product convergence structure.

Proposition 1.1. (see [2], page 26): *Let X, Y, Z be convergence spaces. Then a map $h : X \rightarrow \mathcal{C}(Y, Z)$ is continuous if and only if the associated map*

$$\tilde{h} : X \times Y \rightarrow Z$$

defined by $\tilde{h}(x, y) = h(x)(y)$, is continuous.

Proof: Assume that h is continuous. To show the continuity of $\tilde{h}$, it suffices to take two convergent filters $F \rightarrow x \in X$ and $G \rightarrow y \in Y$, and show that $\tilde{h}(F \times G) \rightarrow \tilde{h}(x, y)$, where the filter $F \times G$ is based on $\{A \times B \mid A \in F, B \in G\}$. Indeed, we have

$$\tilde{h}(F \times G) = h(F)(G) \rightarrow h(x)(y) = \tilde{h}(x, y),$$

since $h(F) \rightarrow h(x)$ by the continuity of h . The converse is proven similarly. ■

Proposition 1.2. (see [2], page 34): *Let X, Y be topological spaces such that X is locally compact and Y is regular. Then the convergence space $\mathcal{C}(X, Y)$ is topologized by the compact-open topology. That is, the convergence relation arising from this topology coincides precisely with the convergence structure given in [Definition 1.7](#).*

2. Curried functional spaces

Proposition 2.1. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k \in \mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ (in the sense that its derived filter converges) if and only if $f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$ whenever $x_k \xrightarrow[k \rightarrow \infty]{} x$ in X .*

Proof: First, let F be the derived filter of $\{f_k\}_{k \in \mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k \in \mathbb{N}}$ converging to $x \in X$, with its derived filter P . Now let V be a neighborhood of $f(x)$ in Y . By the definition of continuous convergence, we have $F(P) \rightarrow f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set

$$B_{n_0, m_0} = \{f_n(x_m) \mid n \geq n_0, m \geq m_0\}.$$

Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.

Conversely, suppose that $f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$ whenever $x_k \xrightarrow[k \rightarrow \infty]{} x \in X$. First let us show that for any subsequence $\{f_{n_k}\}_{k \in \mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$.

To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n \in \mathbb{N}}$ converges to x . Hence we have $f_k(z_k) \xrightarrow[k \rightarrow \infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}(z_{n_k}) \xrightarrow[k \rightarrow \infty]{} f(x). \quad (1)$$

Now, to show that $f_k \rightarrow f$ in $\mathcal{C}(X, Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f . To this end, let P be a filter in X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m \in \mathbb{N}}$ of neighborhoods of x . To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_m}(x_m) \notin V$.

In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P , we see that $x_n \xrightarrow[n \rightarrow \infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n \rightarrow \infty]{} f(x)$ by (1), a contradiction. ■

Definition 2.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$. The *curried space of continuous functions* $C_n(\mathbb{R})$ is defined recursively as follows:

- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$.

Proposition 2.2. For all $n \in \mathbb{N}_0$, the space $C_n(\mathbb{R})$ is a Hausdorff topological vector space.

Proof: If $n = 0$, the statement is trivial. Suppose that $C_n(\mathbb{R})$ is a Hausdorff TVS. We aim to show the same for $C_{n+1}(\mathbb{R})$.

Since $\mathbb{R}$ is locally compact and $C_n(\mathbb{R})$ is regular, we see by [Proposition 1.2](#) that the convergence space $\mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$ is topological. It is given the pointwise vector space structure, i.e.

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x)$$

for $\alpha, \beta \in \mathbb{R}$ and $f, g \in C_{n+1}(\mathbb{R})$.

First of all, as proved in [2] (page 28), we see that $C_{n+1}(\mathbb{R})$ is Hausdorff. It remains to show that the addition and scalar multiplication maps are continuous.

Consider the addition map $+: C_{n+1}(\mathbb{R}) \times C_{n+1}(\mathbb{R}) \rightarrow C_{n+1}(\mathbb{R})$. Let $F_1 \rightarrow f_1$ and $F_2 \rightarrow f_2$ in $C_{n+1}(\mathbb{R})$. We aim to show that $F_1 + F_2 \rightarrow f_1 + f_2$, where $F_1 + F_2$ is a shorthand for $(+)(F_1 \times F_2)$. To this end, let $P \rightarrow x \in \mathbb{R}$. Observe that we have $F_1(P) \rightarrow f_1(x)$ and $F_2(P) \rightarrow f_2(x)$ since $f_i \in \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$. Moreover, since $C_n(\mathbb{R})$ is a TVS, we see that

$$F_1(P) + F_2(P) \rightarrow f_1(x) + f_2(x) = (f_1 + f_2)(x).$$

It remains to show that $(F_1 + F_2)(P)$ is finer than $F_1(P) + F_2(P)$. For $i = 1, 2$, let $C_i \in F_i$ and $A_i \in P$. We then see that

$$C_1(A_1) + C_2(A_2) \supset (C_1 + C_2)(A_1 \cap A_2).$$

Therefore, we have

$$\begin{aligned} F_1(P) + F_2(P) &= [\{C_1(A_1) + C_2(A_2) \mid C_i \in F_i, A_i \in P\}] \\ &\subset [(C_1 + C_2)(A) \mid C_i \in F_i, A \in P] = (F_1 + F_2)(P), \end{aligned}$$

and so $(F_1 + F_2)(P) \rightarrow (f_1 + f_2)(x)$, which allows us to conclude that

$$F_1 + F_2 \rightarrow f_1 + f_2 \in C_{n+1}(\mathbb{R}).$$

Finally, consider the multiplication map $\cdot : \mathbb{R} \times C_{n+1}(\mathbb{R}) \rightarrow C_{n+1}(\mathbb{R})$. Let $P \rightarrow \alpha \in \mathbb{R}$ and $F \rightarrow f \in C_{n+1}(\mathbb{R})$. To show that $P \cdot F \rightarrow \alpha f$, take a filter $Q \rightarrow x \in \mathbb{R}$. One can easily show that $(PF)(Q) = P \cdot F(Q)$, and, since $\cdot : \mathbb{R} \times C_n(\mathbb{R}) \rightarrow C_n(\mathbb{R})$ is continuous, we have

$$(PF)(Q) = P \cdot (F(Q)) \rightarrow \alpha \cdot f(x) = (\alpha f)(x),$$

q.e.d. ■

Proposition 2.3. *Let X be a topological space, $n \in \mathbb{N}_0$. Then a map $h : X \rightarrow C_n(\mathbb{R})$ is continuous if and only if the map*

$$\tilde{h} : X \times \mathbb{R}^n \rightarrow \mathbb{R}$$

defined by $\tilde{h}(x, y_1, y_2, \dots, y_n) = h(x)(y_1)(y_2) \dots (y_n)$, is continuous.

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$, and consider a map $h : X \rightarrow C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$.

By [Proposition 1.1](#), we see that h is continuous if and only if the associated map

$$h' : X \times \mathbb{R} \rightarrow C_n(\mathbb{R})$$

is continuous, which, by the induction hypothesis, is equivalent to the continuity of $\tilde{h}' = \tilde{h}$. ■

Corollary 2.1. *Let $n \geq 1$. Then the correspondence $h \mapsto \tilde{h}$ yields a homeomorphism $C_n(\mathbb{R}) \cong \mathcal{C}(\mathbb{R}^n, \mathbb{R})$.*

Proof: ■

Bibliography

- [1] N. Bourbaki, *General Topology, Part I*. 1966.
- [2] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.